Shop Keeper

Properties

- Customer Presence
- Items in Stock
- Item Retrieved
- Is Alerted

Behaviours

- Idle
- Sell
- Patrol
- Retrieve
- Stock
- Command

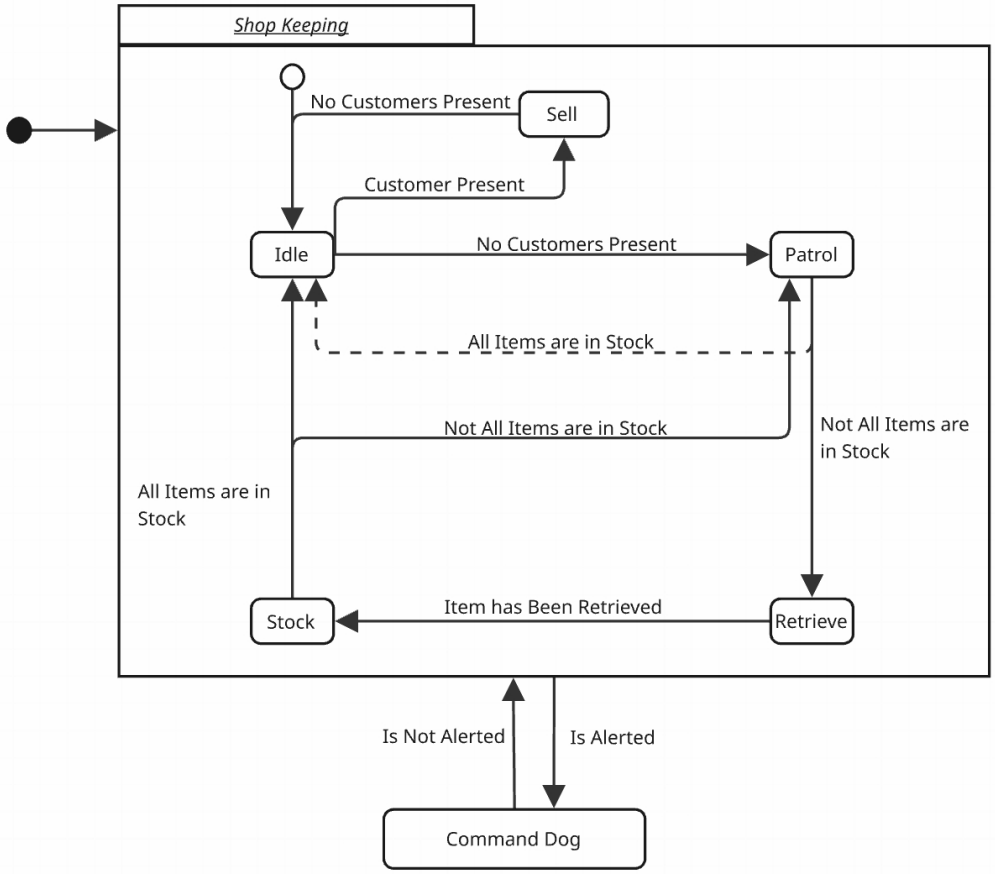
Relationships

- Player-Agent:
  - If shopkeeper agent spots the player pickpocketing or shoplifting, the shopkeeper will command the dog to attack the player.
- Agent-Agent:
  - If a customer alerts the shopkeeper, the shopkeeper will command the dog to attack the player.
  - If the shopkeeper spots the customers shoplifting, the shopkeeper will command the dog to attack the customer.

Feedback

- Idle
  - Shopkeeper will play an Idle animation behind the Till
- Sell
  - Shopkeeper will play a small sound-effect upon selling an Item
- Patrol
  - Shopkeeper will play a hum sound-effect, as well as play an animation of the shopkeeper walking down the store isles
- Retrieve
- Stock
- Command
  - Shopkeeper will play a, relatively, loud sound-effect, commanding the dog.
  - Upon commanding dog, shopkeeper will play a finger-pointing animation
  - Small red exclamation mark will appear above the shopkeeper

Shop Keeper - FSM



Perception

- Vision Cone
  - Can see directly in front of themselves
  - Vision Cone is used to spot:
    - Player shoplifting
    - Player pickpocketing
    - Customer shoplifting

Dog

Properties

- Idle Timer
- Play Timer
- Is Commanded
- Player Interacted

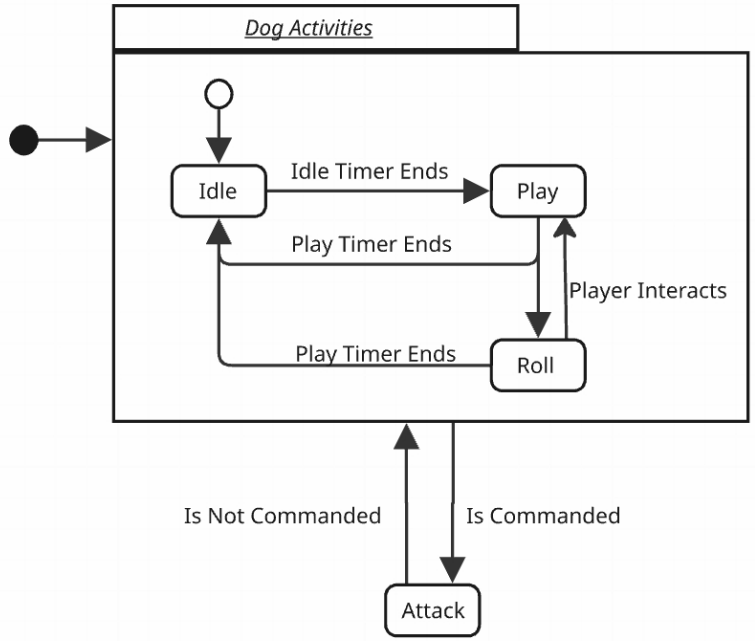
Behaviours

- Idle
- Play
- Roll
- Attack

Relationships

- Player-Agent:
  - If the dog is commanded, the dog will attack the player.
  - If the dog is interacted with by the player during the Play State, the dog will roll-over and allow the player to pet them in the Roll State.
- Agent-Agent:
  - If the shopkeeper commands the dog, the dog will attack the given target (Player or Customer Agent)

Dog - FSM



Feedback

- Idle
  - Dog will play a resting animation during the Idle state
- Play
  - Dog will play bark sound-effects at random
  - Dog will run between areas of the store
- Roll
  - Dog will play an animation of rolling-over and laying on it's back
- Attack
  - Dog will play a gnarl sound-effect
  - Dog will play aggressive bark sound-effects at random
  - Dog will play a bite animation when in range of the target

Perception

- Vision Cone
  - Can see directly in front of themselves
  - Vision Cone is used to guide the Dog towards their target if commanded to Attack
- Interaction Radius
  - Can be interacted with by the Player
- Audio Listener
  - Can hear the command sound effect played by the Shopkeeper

Customer

Properties

- Has Entered Shop
- Player Caught Pickpocketing
- Has Alerted Shopkeeper
- Wanted Items Present
- Has Purchased Items

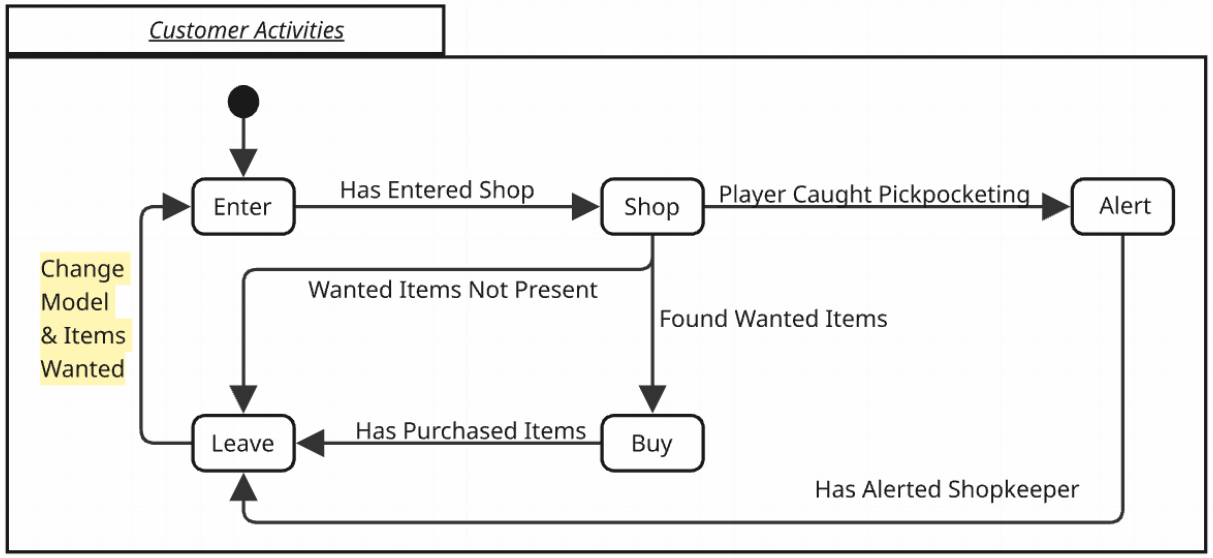
Behaviours

- Enter
- Shop
- Buy
- Leave
- Alert

Relationships

- Player-Agent:
  - Player can pickpocket the Customer Agent
  - Player can be spotted pickpocketing
- Agent-Agent:
  - If Customer spots the Player pickpocketing, the Customer agent will alert the Shopkeeper agent
  - If Customer is spotted shoplifting, the Shopkeeper will be alerted

Customer - FSM



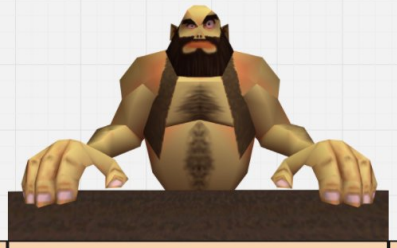
*-The highlighted portion refers to swapping the customers model and their intended wanted items. Swapping these upon the customer leaving the store, allows for reentry without spawning more customer objects

Feedback

- Enter
- Shop
  - Customer will walk to given isles and play a ponder animation
- Buy
  - Customer will play a sound effect upon purchasing their given items
- Leave
- Alert
  - Customer will approach the shopkeeper and play a angry toned mumble sound effect

Perception

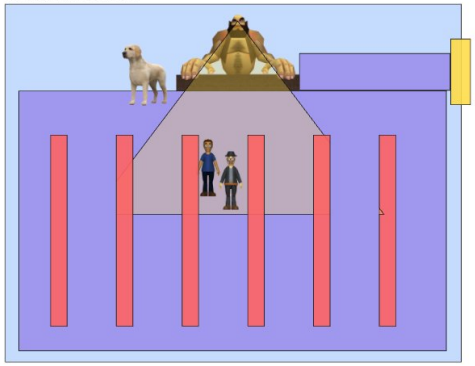
- Vision Cone
  - Can see directly in front of themselves
  - Vision Cone is used to spot:
    - Player shoplifting
    - Player pickpocketing
    - Customer shoplifting
- Interaction Radius
  - Used for being pickpocketed
- Audio Listener
  - Used for detecting Dog if currently targeted
  - Used for Player pickpocketing, if unsuccessful



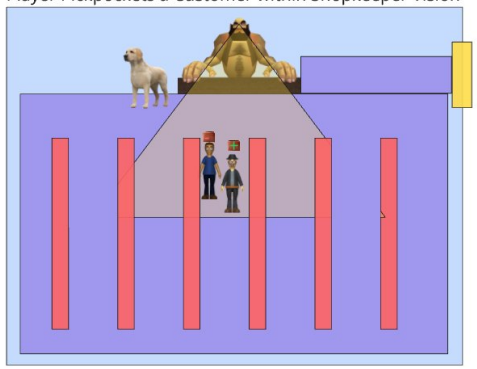
Matrix	Player	Shopkeeper	Dog	Customer
Player	**X**	-Can purchase Items from Shopkeeper	-Can Pet Dog	-Can pickpocket Customer
Shopkeeper	-Can Sell Items to the Player Directly	**X**	-Can Command Dog To Attack	-Can Sell Items to the Customer Directy
Dog	-Can Attack Player	-Can Be Commanded To Attack Target	**X**	- Can Attack Customer
Customer	-Can be pickpocketed by Player	-Can Purchase Items from Shopkeeper	-Can be Attacked by Dog	**X**

Shopkeeper Spots Player Pickpocketing

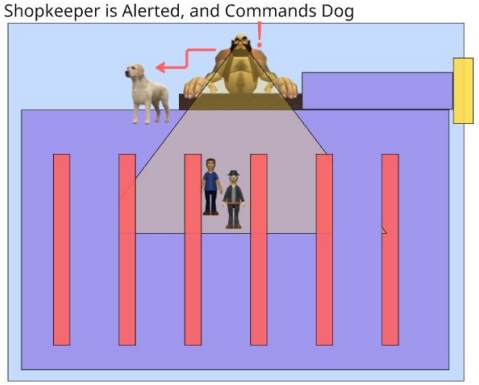
Neutral Position



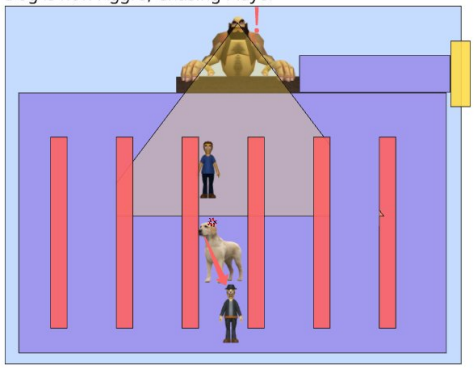
Player Pickpockets a Customer within Shopkeeper Vision



Shopkeeper is Alerted, and Commands Dog

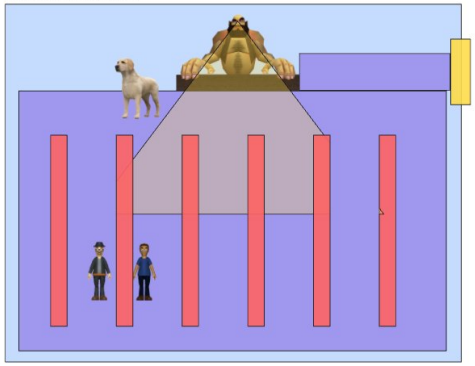


Dog is now Aggro, Chasing Player

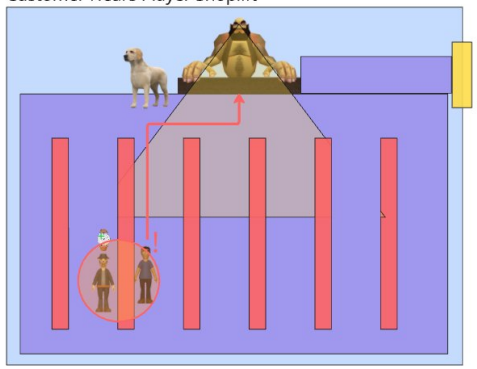


Customer Overhears Player Shoplift

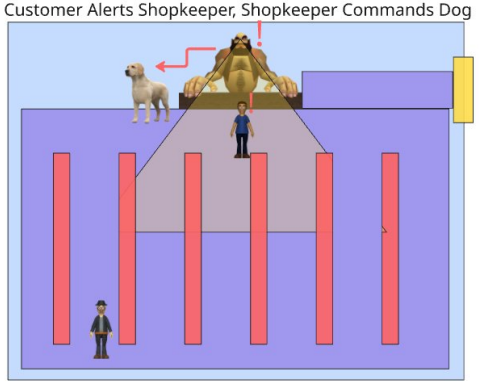
Neutral Position



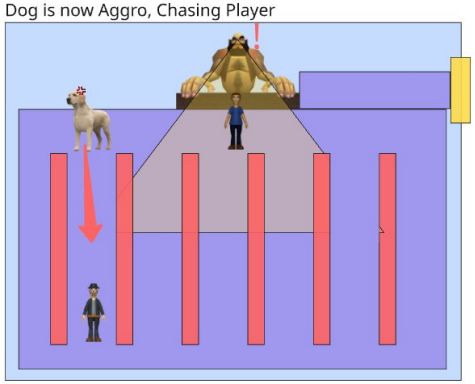
Customer Hears Player Shoplift



Customer Alerts Shopkeeper, Shopkeeper Commands Dog

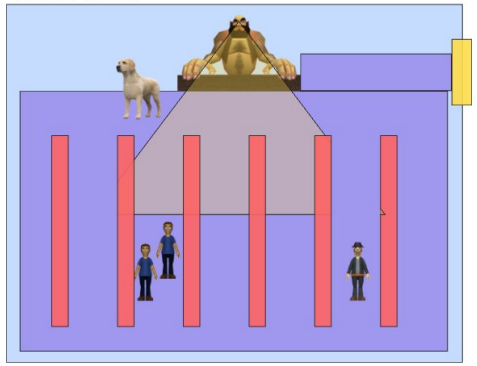


Dog is now Aggro, Chasing Player

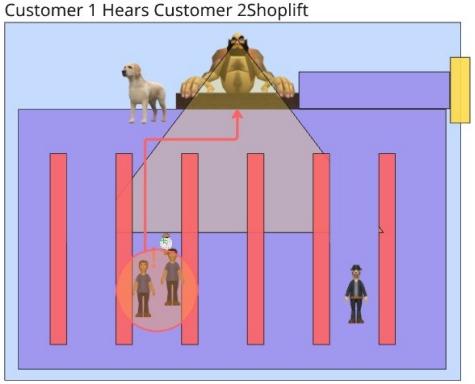


#3. Customer alerts Shopkeeper about another Customer

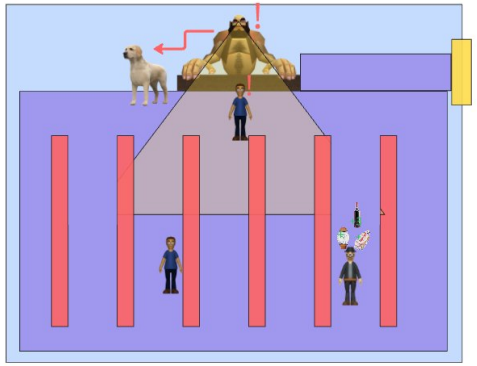
Neutral Position



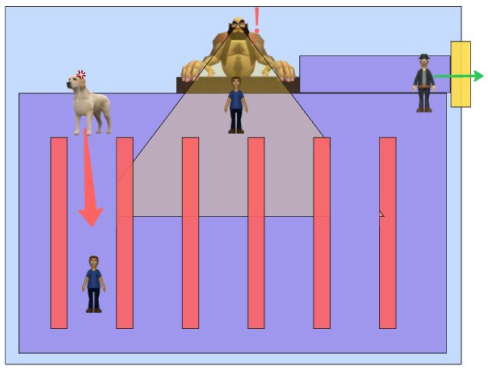
Customer 1 Hears Customer 2Shoplift



Customer 1 Alerts Shopkeeper, Shopkeeper Commands Dog
Player can freely shoplift



Dog is now Aggro, Chasing Customer 2
Player can leave without incident



Design Justification

1. *Justify why you designed these specific AI using the lens of objects, properties, behaviours.*

Before delving into the design justification, I would just like to preface this assignment by saying that these AI's were designed for a non-existing game, in which the player attempts to shoplift without getting caught. The player would have a value quota of stolen goods, and has to successfully steal up to the target value prior to leaving the store.

I designed these AI agents to be relatively simple in their design, allowing the complexity of the system to emerge naturally. For the Shopkeeper, they keep track of items within the shop, and stock them when appropriate and not actively serving a customer. Regarding the Customer, the Customer has a list of intended items to purchase, as well as a chance of shoplifting, should an Item be present that the customer intends to retrieve, the chance of shoplifting will be checked and the Customer will perform the action accordingly. The Dog is relatively simple, following a loop of resting and playing, allowing the player to interact with them passively. Should the Dog be commanded to attack, the dog will stop all other activities to proceed with attacking the given target.

2. *Justify why you designed these specific player-agent and agent-agent relationships.*

These AI agents were designed to compliment each other, providing the opportunity for emergent gameplay. As these individual FSM's can all interact with each other, situations such as Scenario #3, show opportunities for the Player to gather multiple items with less risk of alerting the shopkeeper, due to a Customer overhearing another Customer shoplift, resulting in the Dog targeting a Customer, freeing the player to act freely.

Alongside this, the interactivity between the agents allow the game to feel as if it progresses regardless of the players existence, ergo, the shop feels lively, and that customers will shoplift and the shopkeeper will keep up stock and attempt to prevent shoplifting. In terms of the specific interactions themselves, I designed them with the idea of Customers being random people with random moral values and goals, such as the player.

3. *Justify why you designed these specific feedback states.*

I designed these specific feedback states, as I believe they are necessary for clearly communicating the events and transitions of the AI agents. Audio and visual effects such as animation, exclamation markers, etc. are all pivotal aspects of notifying the player about the events of the game. Alongside this, specific feedback, such as the Shopkeeper commanding the Dog is entirely run off of the Audio Clip being played.